CONVERGED THERMODYNAMIC & DESIGN SUMMARY (EQUAL AREA A1 = A2 = A3 = 68.30 M²)

EQUIPMENT / UNIT	NAOH (WT%)	BPE (°C)	T SOLN (°C)	TSAT,V (°C)	ΔT (°C)	U (W/M²K)	AREA (M²)	DUTY (KW)
EFFECT 1 (E-101)	19.63 %	10.10	143.41	133.31	15.44	3,900	68.30	4,115
EFFECT 2 (E-102)	28.32 %	15.00	130.78	115.78	12.63	2,700	68.30	2,330
EFFECT 3 (E-103)	**50.00 %**	40.00	111.60	71.60	19.18	1,800	68.30	2,360
CONDENSER (E-104)	—	—	—	71.60	—	—	—	2,360
OVERALL TOTALS	**50.00 %**	**65.10**	**—**	**—**	**47.25**	**—**	**204.9 M²**	**8,805 KW**

VAPOR V3: V3 = 3,825 KG/H | TSAT = 71.60 °C (TO SURFACE CONDENSER E-104)

VAPOR V2: V2 = 3,911 KG/H | TSAT = 115.78 °C | Λ2 = 2,172 KJ/KG (TO E-103)

VAPOR V1: V1 = 3,930 KG/H | TSAT = 133.31 °C | Λ1 = 2,134 KJ/KG (TO E-102)

RAW FEED (NAOH SOLN)
F = 400 TONNE/DAY
F = 16,666.67 KG/H
X_F = 15.0 WT% NAOH
T_F = 35.0 °C (PREHEATED)

FIC 101

MOTIVE STEAM SUPPLY
P_S = 6.0 BAR (ABS) [SAT.]
T_S = 158.85 °C
S = 7,105 KG/H (170.5 T/D)
Λ_S = 2,085.26 KJ/KG

FCV 101

CAD SYMBOL & LINE LEGEND
- PRIMARY PROCESS / STEAM LINE
- EVAPORATED VAPOR OVERHEAD LINE
- LIQUID TRANSFER / PRODUCT LINE
- CONDENSATE RECOVERY HEADER
- UTILITY / COOLING WATER LINE
- CENTRIFUGAL PROCESS PUMP
- MANUAL / CONTROL VALVE

E-101
EFFECT 1 (1ST EFFECT)
PI 101 | TI 101
WIRE MESH DEMISTER
DESIGN & OPERATING DATA:
T1 = 143.41 °C (BOILING TEMP)
X1 = 19.63 WT% NAOH
BPE1 = 10.10 °C (DUHRING)
TSAT,1 = 133.31 °C
ΔT1 = 15.44 °C (DRIVING FORCE)
U1 = 3900 W/M²·K
A1 = 68.30 M² (HEAT AREA)
Q1 = 4,115 KW
NORMAL LIQUID LEVEL
BODY H = 8.20 M
CHEST H = 4.20 M
CALANDRIA TUBE BUNDLE (VERTICAL)
VESSEL ID = 2.40 M (TYP)

E-102
EFFECT 2 (2ND EFFECT)
PI 102 | TI 102
WIRE MESH DEMISTER
DESIGN & OPERATING DATA:
T2 = 130.78 °C (BOILING TEMP)
X2 = 28.32 WT% NAOH
BPE2 = 15.00 °C (DUHRING)
TSAT,2 = 115.78 °C
ΔT2 = 12.63 °C (DRIVING FORCE)
U2 = 2700 W/M²·K
A2 = 68.30 M² (HEAT AREA)
Q2 = 2,330 KW
NORMAL LIQUID LEVEL
CALANDRIA TUBE BUNDLE (VERTICAL)

E-103
EFFECT 3 (3RD EFFECT)
PI 103 | TI 103
WIRE MESH DEMISTER
DESIGN & OPERATING DATA:
T3 = 111.60 °C (BOILING TEMP)
X3 = 50.00 WT% NAOH (TARGET)
BPE3 = 40.00 °C (DUHRING)
TSAT,3 = 71.60 °C
ΔT3 = 19.18 °C (DRIVING FORCE)
U3 = 1800 W/M²·K
A3 = 68.30 M² (HEAT AREA)
Q3 = 2,360 KW
NORMAL LIQUID LEVEL
CALANDRIA TUBE BUNDLE (VERTICAL)

E-104
SURFACE CONDENSER
SHELL & TUBE CONDENSER
P3 = 33.6 KPA(ABS) [20 IN.HG VAC]
TSAT = 71.60 °C
QC = 2,360 KW

TO STEAM JET EJECTOR / VACUUM SYSTEM (20 IN.HG)

COOLING WATER OUT
T_CW,OUT = 35.0 °C
ΔT_CW = 10.0 °C

COOLING WATER IN
T_CW,IN = 25.0 °C
W_C = 203 TONNE/H

BAROMETRIC HOTWELL
CONDENSATE SEAL TANK

CALANDRIA CONDENSATE COLLECTION HEADER → STEAM TRAP / BOILER FEED RETURN

P-101
LIQUOR L1: L1 = 12,737 KG/H | X1 = 19.63 WT% NAOH | T1 = 143.41 °C

P-102
LIQUOR L2: L2 = 8,826 KG/H | X2 = 28.32 WT% NAOH | T2 = 130.78 °C

P-103
CONCENTRATED PRODUCT STREAM (P = 5,000 KG/H, 50 WT% NAOH)

FINAL PRODUCT (50 WT% NAOH)
P = 5,000.00 KG/H (120 TONNE/DAY)
X_P = 50.0 WT% NAOH | T3 = 111.60 °C

THIRD ANGLE PROJECTION

TRIPLE-EFFECT FORWARD-FEED EVAPORATOR SYSTEM — NAOH CONCENTRATION (50 WT%)

DRAWING TITLE **PROCESS FLOW DIAGRAM (PFD)**	SUBJECT **PROCESS HEAT TRANSFER (PHT)**	COURSE / BRANCH **B.E. CHEMICAL ENGINEERING**	SCALE **DIMENSIONALLY PROPORTIONED**
STUDENT NAME **SOURISH SENAPATI**	EXAM ROLL NUMBER **002310301035**	ASSIGNED ROLL (LAST 2 DIGITS) **35 (TARGET: 50.0 WT% NAOH)**	DATE **01 / 09 / 2026**
FEED CONDITIONS **400 T/D (16,666.67 KG/H) @ 15.0 WT%, 35°C**	PRODUCT CONDITIONS **120 T/D (5,000.00 KG/H) @ 50.0 WT%, 111.6°C**	MOTIVE STEAM SUPPLY **6.0 BAR(A) SAT. (TS = 158.85°C), S = 7,105 KG/H**	HEAT TRANSFER AREA **A1 = A2 = A3 = 68.30 M² (EQUAL AREA)**

TOTAL EVAPORATION: 11,666.67 KG/H (280 T/D) | STEAM CONSUMPTION: 7,105 KG/H (170.5 T/D) | STEAM ECONOMY: 1.64 | COOLING WATER: 203 T/H | VACUUM: 20 IN.HG (33.6 KPA ABS)